CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – IV Examination – DEC 2011

PT208 BIOSTATISTICS

Date: 06.12.11, Tuesday **Time: 10.00 a.m to10.10 a.m** **Maximum Marks: 5**

Instructions:

1. There are two parts in the paper. **PART –I** contains 5 Questions and **PART – II** contains 2 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 10 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 10 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

1. Standard deviation is the _______ of variance.

- **A.** Square.
- **B.** Half.
- **C.** +ve square root.
- **D.** None of the above.

2. Average Systolic blood pressure is

- **A.** Discrete variable
- **B.** Continuous variable
- **C.** None of (A) and (B)
- **D.** Both of (A) and (B)

3. Which of the following is a unit less measure of dispersion?

- **A.** Standard Deviation
- **B.** Range
- **C.** Coefficient of Variation
- **D.** Mean Deviation

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Subject ______ No. ________
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Section ________
Date ______
Sr. Supervisor's Sign.

4. How many parameters does Normal distribution have?

A. 0

B. 1

C. 2

D. 3

5. If X ~ B (10, 0.75), then P(X = r) =

A. $\binom{10}{r} 0.75^{r} \times 0.25^{10-r}$

B. $\binom{10}{r} \dfrac{0.75^{r}}{(1-0.75)^{10-r}}$

C. $\binom{10}{r} 0.75^{10-r} \times 0.25^{10}$

D. $\binom{10}{r} \dfrac{(1-0.75)^{10-r}}{0.75^{r}}$

End of Part I

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PT208 BIOSTATISTICS

Date: 06.12.11, Tuesday **Time: 10.10 a.m to11.30 a.m** **Maximum Marks: 30**

Instructions:

1. There are two parts in the paper. **PART –I** contains 5 Questions and **PART – II** contains 2 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 1 Hr 20min.
5. Draw the figure wherever necessary.

PART-II

Q. 1 Attempt **any one** of the following: **10**

(a) The following table shows a distribution of bristle number of *Drosophila.* Calculate the mean, median, Standard deviation, coefficient of variation and mode for the following data

Bristle no. (X)	1	2	3	4	5	6	7	8	9	10
No. of individuals (f)	1	4	5	7	9	12	45	57	25	1

(b) The amount of chemical compound Y, which were dissolved in 100 grams of water at various temperatures, X were recorded as follows:

X (^{o}C)	15	15	30	30	45	45	60	60
Y(grams)	12	10	25	21	31	33	44	39

Find the line of regression of y on x and estimate the amount of chemical that will dissolve in 100 grams of water at 50°C.

Q. 2 Attempt **any four** of the following: **20**

(a) What do you mean by sampling with replacement and without replacement?

(b) Draw the Histogram and give your conclusion for the following data of age and No. of patients in a hospital.

Age	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Patients	20	35	60	80	105	120	60	20

(P.T.O)

(c) Patients with low back pain syndrome (LBS) often have a clinical presentation of postural deviation of a lateral lumbar shift (LLS), which is also known as a sciatic scoliosis. The determination of the presence and direction of this lateral postural deviation is necessary to guide treatment in certain physical therapy management approaches. The purpose of the study was to determine the relationship between direction of lateral lumbar shift (LLS) and (1) the occurrence of symptoms during the side bending movement test and (2) the location of symptoms in patients with low back pain syndrome (LBS). The following tables show (1) the direction of lateral shift and positive side bending test.

Direction of shift	Positive side Bending to right	Positive side Bending to Left
Right	22	17
Left	10	13

Do these data indicate a statistically significant relationship between the direction of the lateral shift and the direction of the positive side-bending test? Use 5% level of significance.

$\chi^2_{0.05,1} = 3.84,\ \chi^2_{0.01,1} = 6.63, \chi^2_{0.025,1} = 5.02$

(d) Explain simple correlation in detail.

(e) A sample of 10 patients with fibromyalgia syndrome (FMS) is drawn randomly from a population of such patients and then assigned to receive treatments designed to alleviate pain. The chosen outcome variable, pain intensity as measured in millimeters on a 10-cm visual analog scale (VAS), is measured before and after treatment, and a pain relief score is thereby obtained (pretest score minus posttest score) as

1	11	9	8	11	8	7	10	7	8

Compute 95% confidence limits for the true mean pain relief score.

$t_{0.025,9} = 2.262, t_{0.05,9} = 1.833$

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